No.

Date:

Tugas 5

1. Diket matriks-matriks sebagai berikut

$$A=\begin{bmatrix}2 & 4\\ 5 & 1\end{bmatrix} \quad B=\begin{bmatrix}14 & 4\\ -3 & -5\end{bmatrix} \quad C=\begin{bmatrix}22\\ 12\end{bmatrix}$$

tentukan

a. $2A+3B$

$$=2\begin{bmatrix}2 & 4\\ 5 & 1\end{bmatrix}+3\begin{bmatrix}14 & 4\\ -3 & -5\end{bmatrix}$$

$$=\begin{bmatrix}4 & 8\\ 10 & 2\end{bmatrix}+\begin{bmatrix}42 & 12\\ -9 & -15\end{bmatrix}$$

$$=\begin{bmatrix}46 & 20\\ 1 & -13\end{bmatrix}$$

b. $2A^T+3B^T$

$$=2\begin{bmatrix}2 & 5\\ 4 & 1\end{bmatrix}+3\begin{bmatrix}14 & -3\\ 4 & -5\end{bmatrix}$$

$$=\begin{bmatrix}4 & 10\\ 8 & 2\end{bmatrix}+\begin{bmatrix}42 & -9\\ 12 & -15\end{bmatrix}$$

$$=\begin{bmatrix}46 & 1\\ 20 & -13\end{bmatrix}$$

C. $A \cdot C$

$$=\begin{bmatrix}2 & 4\\ 5 & 1\end{bmatrix}\begin{bmatrix}22\\ 12\end{bmatrix}$$

$$=\begin{bmatrix}44+48\\ 110+12\end{bmatrix} \Rightarrow \begin{bmatrix}92\\ 122\end{bmatrix}$$

2. Diketahui Matriks

$$A=\begin{bmatrix}0 & -2 & 1\\ 4 & -1 & 0\end{bmatrix} \quad B=\begin{bmatrix}1 & 4\\ 0 & 1\\ 4 & -1\end{bmatrix}$$

$$A \cdot B=\begin{bmatrix}0\times1+(-2)\times0+1\times4 & 0\times4+(-2)(1)+1(-1)\\ 4\times1+(-1)(0)+0\times4 & 4(4)+(-1)(1)+0(-1)\end{bmatrix}=\begin{bmatrix}4 & -3\\ 4 & 15\end{bmatrix}$$

BigBOSS

No.

Date:

3 Tentukan det dari matriks dengan metode sarus

a $A = \begin{pmatrix} 2 & 6 \\ 5 & -4 \end{pmatrix}$

$\det(A) = (2 \times -4) - (6 \times 5)$

$= -8 - 30$

$= -38$

b $B = \begin{pmatrix} 1 & 3 & 0 \\ -2 & 5 & 8 \\ 4 & -1 & -3 \end{pmatrix}$

$= \begin{vmatrix} 1 & 3 & 0 & 1 & 3 \\ -2 & 5 & 8 & -2 & 5 \\ 4 & -1 & -3 & 4 & -1 \end{vmatrix}$

$= (1 \times 5 \times (-3) + 3 \times 8 \times 4 + 0 \times (-2) \times (-1)) - [0 \times 5 \times 4 + 1 \times 8 \times (-1) + 3 \times (-2) \times (-3)]$

$= (-15 + 96 + 0) - (0 - 8 + 18)$

$= 81 - 10$

$= 71$

c $C = \begin{pmatrix} -10 & 3 & -4 \\ 3 & 10 & 9 \\ -10 & 2 & 5 \end{pmatrix}$

$= \begin{vmatrix} -10 & 3 & -4 & -10 & 3 \\ -3 & 10 & 9 & 3 & 10 \\ -10 & 2 & 5 & -10 & 2 \end{vmatrix}$

$= (-10 \times 10 \times 5 + 3 \times 9 \times (-10) + (-4) \times 3 \times 2) - (-4 \times 10 \times (-10) + (-10) \times 9 \times 2 + 3 \times 3 \times 5)$

$= (-500 - 270 - 24) - (400 - 180 + 45)$

$= -794 - 265$

$= -1059$

No.

Date:

4. Kofaktor dari matriks

a. $K = \begin{pmatrix} -2 & 2 & -8 \\ 8 & 9 & 9 \\ 0 & -1 & -4 \end{pmatrix}$

• Minor Elemen matriks

$M_{11} = \begin{vmatrix} 9 & 9 \\ -1 & -4 \end{vmatrix} = -36+9 = -27$ $M_{12} = \begin{vmatrix} 8 & 9 \\ 0 & -4 \end{vmatrix} = -32-0 = -32$ $M_{13} = \begin{vmatrix} 8 & 9 \\ 0 & -1 \end{vmatrix} = -8-0 = -8$

$M_{21} = \begin{vmatrix} 2 & -8 \\ -1 & -4 \end{vmatrix} = -8-8 = -16$ $M_{22} = \begin{vmatrix} -2 & -8 \\ 0 & -4 \end{vmatrix} = 8-0 = 8$ $M_{23} = \begin{vmatrix} -2 & 2 \\ 0 & -1 \end{vmatrix} = 2-0 = 2$

$M_{31} = \begin{vmatrix} 2 & -8 \\ 9 & 9 \end{vmatrix} = 18+72 = 90$ $M_{32} = \begin{vmatrix} -2 & -8 \\ 8 & 9 \end{vmatrix} = -8+64 = 46$ $M_{33} = \begin{vmatrix} -2 & 2 \\ 8 & 9 \end{vmatrix} = -8-16 = -24$

• Kofaktor

$kof(k) = \begin{pmatrix} M_{11} & -M_{12} & M_{13} \\ -M_{21} & M_{22} & -M_{23} \\ M_{31} & -M_{32} & M_{33} \end{pmatrix} = \begin{pmatrix} -27 & -(-32) & -8 \\ -(16) & 8 & -2 \\ 90 & -46 & -24 \end{pmatrix} = \begin{pmatrix} -27 & 32 & -8 \\ 16 & 8 & -2 \\ 90 & -46 & -24 \end{pmatrix}$

• Adjoint nya

$Adj(k) = (kof(k))^T$

$= \begin{pmatrix} -27 & 32 & -8 \\ 16 & 8 & -2 \\ 90 & -46 & -24 \end{pmatrix}^T \Rightarrow \begin{pmatrix} -27 & 16 & 90 \\ 32 & 8 & -46 \\ -8 & -2 & -24 \end{pmatrix}$

BIG BOSS

No.

Date:

b) $L = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 2 & -1 \\ 1 & 1 & 3 \end{pmatrix}$

- Minor elemt matrik L

$M_{11} = \begin{vmatrix} 2 & -1 \\ 1 & 3 \end{vmatrix} = 6+1 = 7$ $\quad M_{12} = \begin{vmatrix} 1 & -1 \\ 1 & 3 \end{vmatrix} = 3+1 = 4$ $\quad M_{13} = \begin{vmatrix} 1 & 2 \\ 1 & 1 \end{vmatrix} = 1-2 = -1$

$M_{21} = \begin{vmatrix} 0 & 1 \\ 1 & 3 \end{vmatrix} = 0-1 = -1$ $\quad M_{22} = \begin{vmatrix} 1 & 1 \\ 1 & 3 \end{vmatrix} = 3-1 = 2$ $\quad M_{23} = \begin{vmatrix} 1 & 0 \\ 1 & 1 \end{vmatrix} = 1-0 = 1$

$M_{31} = \begin{vmatrix} 0 & 1 \\ 2 & -1 \end{vmatrix} = 0-2 = -2$ $\quad M_{32} = \begin{vmatrix} 1 & 1 \\ 1 & -1 \end{vmatrix} = -1-1 = -2$ $\quad M_{33} = \begin{vmatrix} 1 & 0 \\ 1 & 2 \end{vmatrix} = 2-0 = 2$

- kofaktor

$$kof(L) = \begin{pmatrix} M_{11} & -M_{12} & M_{13} \\ -M_{21} & M_{22} & -M_{23} \\ M_{31} & -M_{32} & M_{33} \end{pmatrix} = \begin{pmatrix} +7 & -4 & +(-1) \\ -(-1) & +2 & -1 \\ +(-2) & -(-2) & +2 \end{pmatrix} \Rightarrow \begin{pmatrix} 7 & -4 & -1 \\ 1 & 2 & -1 \\ -2 & 2 & 2 \end{pmatrix}$$

- Adjointnya

$$Adj(L) = (kof(L))^T$$

$$= \begin{pmatrix} 7 & -4 & -1 \\ 1 & 2 & -1 \\ -2 & 2 & 2 \end{pmatrix}^T \Rightarrow \begin{pmatrix} 7 & 1 & -2 \\ -4 & 2 & 2 \\ -1 & -1 & 2 \end{pmatrix}$$

BIS BOSS

No.

Date:

c. $M = \begin{pmatrix} -10 & 3 & -4 \\ 3 & 10 & 9 \\ -10 & 2 & 5 \end{pmatrix}$

- Minor elemen matriks

$M_{11} = \begin{vmatrix} 10 & 9 \\ 2 & 5 \end{vmatrix} = 50 - 18 = 32$ $\quad M_{12} = \begin{vmatrix} 3 & 9 \\ -10 & 5 \end{vmatrix} = 15 + 90 = 105$ $\quad M_{13} = \begin{vmatrix} 3 & 10 \\ -10 & 2 \end{vmatrix} = 6 + 100 = 106$

$M_{21} = \begin{vmatrix} 3 & -4 \\ 2 & 5 \end{vmatrix} = 15 + 8 = 23$ $\quad M_{22} = \begin{vmatrix} -10 & -4 \\ -10 & 5 \end{vmatrix} = -50 - 40 = -90$ $\quad M_{23} = \begin{vmatrix} -10 & 3 \\ -10 & 2 \end{vmatrix} = -20 + 30 = 10$

$M_{31} = \begin{vmatrix} 3 & -4 \\ 10 & 9 \end{vmatrix} = 27 + 40 = 67$ $\quad M_{32} = \begin{vmatrix} -10 & -4 \\ 3 & 9 \end{vmatrix} = -90 + 12 = -78$ $\quad M_{33} = \begin{vmatrix} -10 & 3 \\ 3 & 10 \end{vmatrix} = -100 - 9 = -109$

- Kofaktor

$$kof(M) = \begin{pmatrix} M_{11} & -M_{12} & M_{13} \\ -M_{21} & M_{22} & -M_{23} \\ M_{31} & -M_{32} & M_{33} \end{pmatrix} = \begin{pmatrix} +32 & -105 & +106 \\ -23 & +(-90) & -10 \\ +67 & -(-78) & +(-109) \end{pmatrix} = \begin{pmatrix} 32 & -105 & 106 \\ -23 & -90 & -10 \\ 67 & 78 & -109 \end{pmatrix}$$

- $Adj(M) = (kof(k))^T$

$$= \begin{pmatrix} 32 & -105 & 106 \\ -23 & -90 & -10 \\ 67 & 78 & -109 \end{pmatrix}^T \Rightarrow \begin{pmatrix} 32 & -23 & 67 \\ -105 & -90 & 78 \\ 106 & -10 & -109 \end{pmatrix}$$

BIG BOSS

No.

Date:

5. Tentukan invers dari matriks

a. $\begin{pmatrix} 3 & 6 \\ 5 & -4 \end{pmatrix}^{-1} = \frac{1}{det} \begin{pmatrix} -4 & -6 \\ -5 & 3 \end{pmatrix}$

$= \frac{1}{-12-30} \begin{pmatrix} -4 & -6 \\ -5 & 3 \end{pmatrix}$

$= \frac{1}{-42} \begin{pmatrix} -4 & -6 \\ -5 & 3 \end{pmatrix}$

$= -\frac{1}{42} \begin{pmatrix} -4 & -6 \\ -5 & 3 \end{pmatrix}$

$= \begin{pmatrix} \frac{2}{21} & \frac{1}{7} \\ \frac{5}{42} & -\frac{1}{14} \end{pmatrix}$

b. $\begin{vmatrix} 1 & 0 & 1 \\ 1 & 2 & -1 \\ 1 & 1 & 3 \end{vmatrix}^{-1}$ = • Cari determinan dulu

$\begin{vmatrix} 1 & 0 & 1 & 1 & 0 \\ 1 & 2 & -1 & 1 & 2 \\ 1 & 1 & 3 & 1 & 1 \end{vmatrix} = (6+0+1)-(2-1+0)$

$= 7 - 1$

$= 6$

• Mencari adj dulu

$Adj = (kofaktor)^T$

$= \begin{pmatrix} M_{11} & -M_{12} & M_{13} \\ -M_{21} & M_{22} & -M_{23} \\ M_{31} & -M_{32} & M_{33} \end{pmatrix}^T \Rightarrow \begin{pmatrix} +(7) & -(4) & +(-1) \\ -(-1) & 2 & -(1) \\ -(2) & -(-2) & 2 \end{pmatrix}^T$

$= \begin{pmatrix} 7 & -4 & -1 \\ 1 & 2 & -1 \\ -2 & 2 & 2 \end{pmatrix}^T = \begin{pmatrix} 7 & 1 & -2 \\ -4 & 2 & 2 \\ -1 & -1 & 2 \end{pmatrix}$

• Inversnya $= \frac{adj}{det} \Rightarrow \frac{1}{6} \begin{pmatrix} 7 & 1 & -2 \\ -4 & 2 & 2 \\ -1 & -1 & 2 \end{pmatrix} \Rightarrow \begin{pmatrix} \frac{7}{6} & \frac{1}{6} & -\frac{1}{3} \\ -\frac{2}{3} & \frac{1}{3} & \frac{1}{3} \\ -\frac{1}{6} & -\frac{1}{6} & \frac{1}{3} \end{pmatrix}$

BIG BOSS

No.

Date:

C. $\begin{vmatrix} 3 & 2 & 1 \\ 1 & 1 & -1 \\ 5 & 1 & -1 \end{vmatrix}$ • cari determinan $= \begin{vmatrix} 3 & 2 & 1 & 3 & 2 \\ 1 & 1 & -1 & 1 & 1 \\ 5 & 1 & -1 & 5 & 1 \end{vmatrix} = (-3-10+1)-(5-3-2)$

$= -12-0$

$= -12$

• Cari Adj nya

$Adj = (kofaktor(C))^T$

$= \begin{pmatrix} M_{11} & -M_{12} & M_{13} \\ -M_{21} & M_{22} & -M_{23} \\ M_{31} & -M_{32} & M_{33} \end{pmatrix}^T = \begin{pmatrix} +(0) & -(-4) & +(-4) \\ -(-3) & +(-8) & -(-7) \\ +(-3) & -(-4) & +(1) \end{pmatrix}^T$

$\Rightarrow \begin{pmatrix} 0 & -4 & -4 \\ 3 & -8 & 7 \\ -3 & 4 & 1 \end{pmatrix}^T \Rightarrow \begin{pmatrix} 0 & 3 & -3 \\ -4 & -8 & 4 \\ -4 & 7 & 1 \end{pmatrix}$

• Inversnya $= \frac{Adj}{det} = \frac{1}{-12} \begin{pmatrix} 0 & 3 & -3 \\ -4 & -8 & 4 \\ -4 & 7 & 1 \end{pmatrix}$

$\Rightarrow \begin{pmatrix} 0 & -\frac{1}{4} & \frac{1}{4} \\ \frac{1}{3} & \frac{2}{3} & -\frac{1}{3} \\ \frac{1}{3} & -\frac{7}{12} & -\frac{1}{12} \end{pmatrix}$

B. // Selesaikan SPL berikut dengan aturan Cramer

a. $x-2y=6$

$2x+2y=-3$ $\Rightarrow \begin{pmatrix} 1 & -2 \\ 2 & 2 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 6 \\ -3 \end{pmatrix}$

$D = \begin{vmatrix} 1 & -2 \\ 2 & 2 \end{vmatrix} = 2-(-4) = 6$ $\quad D_x = \begin{vmatrix} 6 & -2 \\ -3 & 2 \end{vmatrix} = 12-6 = 6$ $\quad D_y = \begin{vmatrix} 1 & 6 \\ 2 & -3 \end{vmatrix} = -3-12 = -15$

Jadi, $x = \frac{D_x}{D} = \frac{6}{6} = 1$ // dan $y = \frac{D_y}{D} = \frac{-15}{6} = -\frac{5}{2}$ //

$(x,y) = (1, -\frac{5}{2})$

BigBOSS

No.

Date:

b. $-0{,}5x_1 + 0{,}4x_2 = 0{,}7$

$1{,}2x_1 - 0{,}3x_2 = 3{,}6$

$\Rightarrow \begin{pmatrix} -0{,}5 & 0{,}4 \\ 1{,}2 & -0{,}3 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0{,}7 \\ 3{,}6 \end{pmatrix}$

$D = \begin{vmatrix} -0{,}5 & 0{,}4 \\ 1{,}2 & -0{,}3 \end{vmatrix} = 0{,}15 - 0{,}48 = -0{,}33$

$D_x = \begin{vmatrix} 0{,}7 & 0{,}4 \\ 3{,}6 & -0{,}3 \end{vmatrix} = -0{,}21 - 1{,}44 = -1{,}65$

$D_y = \begin{vmatrix} -0{,}5 & 0{,}7 \\ 1{,}2 & 3{,}6 \end{vmatrix} = -1{,}8 - 0{,}84 = -2{,}64$

Jadi, x nya $\frac{D_x}{D} = \frac{-1{,}65}{-0{,}33} = 5$ dan y nya $= \frac{D_y}{D} = \frac{-2{,}64}{-0{,}33} = 8$; $(x, y) = (5, 8)$

2. Selesaikan SPL berikut dgn aturan Cramer:

a. $2x + 3y + z = -11$

$2x - 3y + z = -17$

$2x + 3y - z = -3$

$\Rightarrow \begin{vmatrix} 2 & 3 & 1 \\ 2 & -3 & 1 \\ 2 & 3 & -1 \end{vmatrix} \begin{vmatrix} x \\ y \\ z \end{vmatrix} = \begin{vmatrix} -11 \\ -17 \\ -3 \end{vmatrix}$

$D = \begin{vmatrix} 2 & 3 & 1 & 2 & 3 \\ 2 & -3 & 1 & 2 & -3 \\ 2 & 3 & -1 & 2 & 3 \end{vmatrix} = (6 + 6 + 6) - (-6 + 6 - 6)$

$= 18 + 6$

$= 24$

$D_x = \begin{vmatrix} -11 & 3 & 1 & -11 & 3 \\ -17 & -3 & 1 & -17 & -3 \\ -3 & 3 & -1 & -3 & 3 \end{vmatrix} \Rightarrow (-33 - 9 - 51) - (9 - 33 + 51) \Rightarrow -93 - 27$

$\Rightarrow -120$

$D_y = \begin{vmatrix} 2 & -11 & 1 & 2 & -11 \\ 2 & -17 & 1 & 2 & -17 \\ 2 & -3 & -1 & 2 & -3 \end{vmatrix} \Rightarrow (34 - 22 - 6) - (-34 - 6 + 22) \Rightarrow 6 - (-18)$

$\Rightarrow 24$

$D_z = \begin{vmatrix} 2 & 3 & -11 & 2 & 3 \\ 2 & -3 & -17 & 2 & -3 \\ 2 & 3 & -3 & 2 & 3 \end{vmatrix} \Rightarrow (18 - 102 - 66) - (66 - 102 - 18) \Rightarrow -150 - (-54)$

$= -96$

Jadi, $x = \frac{D_x}{D} = \frac{-120}{24} = -5$; $y = \frac{D_y}{D} = \frac{24}{24} = 1$; $z = \frac{D_z}{D} = \frac{-96}{24} = -4$

$(x, y, z) = (-5, 1, -4)$

BigBOSS

No.

Date:

b $x_1 + 2x_2 + 3x_3 = 5$

$3x_1 - x_2 + 2x_3 = 8$ $\Rightarrow \begin{pmatrix} 1 & 2 & 3 \\ 3 & -1 & 2 \\ 4 & -6 & -4 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 5 \\ 8 \\ -2 \end{pmatrix}$

$4x_1 - 6x_2 - 4x_3 = -2$

$D = \begin{vmatrix} 1 & 2 & 3 & 1 & 2 \\ 3 & -1 & 2 & 3 & -1 \\ 4 & -6 & -4 & 4 & -6 \end{vmatrix} = (4 + 16 - 54) - (-12 - 12 - 24)$

$= (-34)(-48)$

$= -34 + 48 \Rightarrow 14$ //

$D_{x_1} = \begin{vmatrix} 5 & 2 & 3 & 5 & 2 \\ 8 & -1 & 2 & 8 & -1 \\ -2 & -6 & -4 & -2 & -6 \end{vmatrix} = (20 + (-8) - 144) - (6 - 60 - 64)$

$= (-132) - (-118)$

$= -132 + 118 = -14$ //

$D_{x_2} = \begin{vmatrix} 1 & 5 & 3 & 1 & 5 \\ 3 & 8 & 2 & 3 & 8 \\ 4 & -2 & -4 & 4 & -2 \end{vmatrix} = (-32 + 40 - 18) - (96 - 4 - 60)$

$= (-10) - (32) \Rightarrow -42$ //

$D_{x_3} = \begin{vmatrix} 1 & 2 & 5 & 1 & 2 \\ 3 & -1 & 8 & 3 & -1 \\ 4 & -6 & -2 & 4 & -6 \end{vmatrix} = (2 + 64 - 90) - (-20 - 48 - 12) \Rightarrow (-24) - (-80)$

$\Rightarrow 56$

jadi, x_1 adalah $\frac{D_{x_1}}{D} = \frac{-14}{14} = -1$; x_2 adalah $\frac{D_{x_2}}{D} = \frac{-42}{14} = -3$; x_3 adalah $\frac{D_{x_3}}{D} = \frac{56}{14} = 4$

$(x_1, x_2, x_3) = (-1, -3, 4)$

3. Selesaikan SPL berikut dengan menggunakan invers matriks

a) $\frac{x}{8} - y = \frac{5}{2}$

$3x + \frac{y}{3} = 13$ $\Rightarrow \begin{pmatrix} \frac{1}{8} & -1 \\ 3 & \frac{1}{3} \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 5/2 \\ 13 \end{pmatrix} \Rightarrow \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \frac{1}{8} & -1 \\ 3 & \frac{1}{3} \end{pmatrix}^{-1} \begin{pmatrix} 5/2 \\ 13 \end{pmatrix}$

- Cari inversnya

$A = \begin{pmatrix} \frac{1}{8} & -1 \\ 3 & \frac{1}{3} \end{pmatrix} \Rightarrow \frac{1}{\left(\frac{1}{8}\cdot\frac{1}{3}\right) - (-3)} \begin{pmatrix} \frac{1}{3} & 1 \\ -3 & \frac{1}{8} \end{pmatrix} = \frac{1}{\frac{73}{24}} \begin{pmatrix} \frac{1}{3} & 1 \\ -3 & \frac{1}{8} \end{pmatrix}$

BIG BOSS

No.

Date:

$= \frac{24}{73}\begin{pmatrix} \frac{1}{3} & 1 \\ -3 & \frac{1}{8} \end{pmatrix} \Rightarrow \begin{pmatrix} \frac{8}{73} & \frac{24}{73} \\ \frac{-72}{73} & \frac{3}{73} \end{pmatrix}$

- Jadi

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 8/73 & 24/73 \\ -72/73 & 3/73 \end{pmatrix} \begin{pmatrix} 5/2 \\ 13 \end{pmatrix}$$

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 8/73 \cdot 5/2 + 24/73 \cdot 13 \\ -72/73 \cdot 5/2 + 3/73 \cdot 13 \end{pmatrix} \Rightarrow \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 332/73 \\ -141/73 \end{pmatrix}$$

jadi $x = 332/73$ dan $y = -141/73$

b $6x_1 - x_2 = 5$

$5x_1 + 2x_2 = 7$ $\Rightarrow \begin{pmatrix} 6 & -1 \\ 5 & 2 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 5 \\ 7 \end{pmatrix}$

$$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 6 & -1 \\ 5 & 2 \end{pmatrix}^{-1}\begin{pmatrix} 5 \\ 7 \end{pmatrix}$$

- Cari invers dari

$A = \begin{pmatrix} 6 & -1 \\ 5 & 2 \end{pmatrix}$ $A^{-1} = \frac{1}{12-(-5)}\begin{pmatrix} 2 & 1 \\ -5 & 6 \end{pmatrix} \Rightarrow \frac{1}{17}\begin{pmatrix} 2 & 1 \\ -5 & 6 \end{pmatrix}$

$= \begin{pmatrix} 2/17 & 1/17 \\ -5/17 & 6/17 \end{pmatrix}$

- Jadi

$\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 2/17 & 1/17 \\ -5/17 & 6/17 \end{pmatrix}\begin{pmatrix} 5 \\ 7 \end{pmatrix} \Rightarrow \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 10/17 + 7/17 \\ -25/17 + \frac{42}{17} \end{pmatrix} \Rightarrow \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

jadi $x_1 = 1$ dan $x_2 = 1$

Big BOSS

No.

Date:

4. Selesaikan SPL Berikut use Invers

a. $4x - y + 2z = 15$
$x + 2y + 3z = 9$
$2x + y - z = 0$

$$\Rightarrow \begin{pmatrix} 4 & -1 & 2 \\ 1 & 2 & 3 \\ 2 & 1 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 15 \\ 9 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 4 & -1 & 2 \\ 1 & 2 & 3 \\ 2 & 1 & -1 \end{pmatrix}^{-1} \begin{pmatrix} 15 \\ 9 \\ 0 \end{pmatrix}$$

- Invers dari

$$A = \begin{pmatrix} 4 & -1 & 2 \\ 1 & 2 & 3 \\ 2 & 1 & -1 \end{pmatrix} \Rightarrow A^{-1} = \frac{\text{Adj}A}{\det A}$$

- Determinan

$$\begin{vmatrix} 4 & -1 & 2 & 4 & -1 \\ 1 & 2 & 3 & 1 & 2 \\ 2 & 1 & -1 & 2 & 1 \end{vmatrix} = (-8 - 6 + 2) - (8 + 12 + 1)$$
$$= -12 - 21$$
$$= -33$$

- Adj A

$$\text{Adj}A = \begin{bmatrix} M_{11} & -M_{12} & M_{13} \\ -M_{21} & M_{22} & -M_{23} \\ M_{31} & -M_{32} & M_{33} \end{bmatrix}^T = \begin{bmatrix} +(-5) & -(-7) & +(-3) \\ -(-1) & +(-8) & -(6) \\ +(-7) & -(10) & +9 \end{bmatrix}^T \Rightarrow \begin{bmatrix} -5 & 7 & -3 \\ 1 & -8 & -6 \\ -7 & -10 & 9 \end{bmatrix}^T$$

$$\Rightarrow \begin{pmatrix} -5 & 1 & -7 \\ 7 & -8 & -10 \\ -3 & -6 & 9 \end{pmatrix}$$

A^{-1}

- $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \left[\frac{\text{Adj}A}{\det A}\right] \begin{bmatrix} 15 \\ 9 \\ 0 \end{bmatrix}$

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = -\frac{1}{33} \begin{pmatrix} -5 & 1 & -7 \\ 7 & -8 & -10 \\ -3 & -6 & 9 \end{pmatrix} \begin{pmatrix} 15 \\ 9 \\ 0 \end{pmatrix}$$

$\Rightarrow x = -\frac{1}{33}(-75 + 9 + 0) \Rightarrow \frac{-66}{-33} \Rightarrow 2$

BigBOSS

Hari: _____

Date: _____

- $y = -\frac{1}{33}(105 - 72 + 0) \Rightarrow \frac{33}{-33} = -1$ //
- $z = -\frac{1}{33}(-45 - 54 + 0) \Rightarrow \frac{-99}{-33} = 3$ //

jadi $(x, y, z) = (2, -1, 3)$

b $x_1 + 3x_2 + 2x_3 = 3$

$2x_1 - x_2 - 3x_3 = -8$

$5x_1 + 2x_2 + x_3 = 9$

$$\Rightarrow \begin{pmatrix} 1 & 3 & 2 \\ 2 & -1 & -3 \\ 5 & 2 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 3 \\ -8 \\ 9 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 & 3 & 2 \\ 2 & -1 & -3 \\ 5 & 2 & 1 \end{pmatrix}^{-1} \begin{pmatrix} 3 \\ -8 \\ 9 \end{pmatrix}$$

- Invers dari

$$A = \begin{pmatrix} 1 & 3 & 2 \\ 2 & -1 & -3 \\ 5 & 2 & 1 \end{pmatrix} \Rightarrow A^{-1} = \frac{\text{Adj } A}{\det A}$$

- determinan

$$\begin{vmatrix} 1 & 3 & 2 & 1 & 3 \\ 2 & -1 & -3 & 2 & -1 \\ 5 & 2 & 1 & 5 & 2 \end{vmatrix} = (-1 - 45 + 8) - (-10 - 6 + 6)$$

$$= -38 - (-10)$$

$$= -38 + 10$$

$$= -28$$

- Adj A

$$\text{Adj } A = \begin{pmatrix} M_{11} & -M_{12} & M_{13} \\ -M_{21} & M_{22} & -M_{23} \\ M_{31} & -M_{32} & M_{33} \end{pmatrix}^T = \begin{pmatrix} +5 & -17 & +9 \\ -(-1) & +(-9) & -(-13) \\ +(-7) & -(7) & +(7) \end{pmatrix}^T = \begin{pmatrix} 5 & -17 & 9 \\ 1 & -9 & 13 \\ -7 & 7 & -7 \end{pmatrix}^T$$

$$= \begin{pmatrix} 5 & 1 & -7 \\ -17 & -9 & 7 \\ 9 & 13 & -7 \end{pmatrix}$$

• $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = A^{-1} \begin{pmatrix} 3 \\ -8 \\ 9 \end{pmatrix}$

$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \frac{\text{Adj } A}{\det A} \begin{pmatrix} 3 \\ -8 \\ 9 \end{pmatrix} \Rightarrow \frac{1}{-28} \begin{pmatrix} 5 & 1 & -7 \\ -17 & -9 & 7 \\ 9 & 17 & -7 \end{pmatrix} \begin{pmatrix} 3 \\ -8 \\ 9 \end{pmatrix}$

• x nya $\Rightarrow \frac{1}{-28}(15 - 8 - 63) = \frac{-56}{-28} = 2$

• y nya $\Rightarrow \frac{1}{-28}(-51 + 72 + 63) = 84 = -3$

• z nya $\Rightarrow \frac{1}{-28}(27 - 104 - 63) = \frac{-140}{-28} = 5$

jadi $(x, y, z) = (2, -3, 5)$: